

Case Study 6 — Detecting an Unusual Customer

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400.

A particular customer spends ₹3,200.

Given Data

- Mean amount spent, $\mu = ₹2000$
- Standard Deviation, $\sigma = ₹400$
- Customer's spending, $X = ₹3200$

1. Calculate the customer's z-score.

Formula $Z = \frac{X - \mu}{\sigma}$

Where,

- X = Customer's spending = ₹3200
- μ = Mean spending = ₹2000
- σ = Standard deviation = ₹400

Substitute the values $Z = \frac{3200 - 2000}{400}$

Calculate the numerator , $3200 - 2000 = 1200$

Now, $Z = \frac{1200}{400}$, $Z = 3$

Answer $Z = 3$

The customer's spending is 3 standard deviations above the average.

2. Is the customer's spending unusually high compared with the average?

A z-score tells us how far a value is from the mean.

General interpretation:

- $Z = 0 \rightarrow$ Exactly at the mean.
- $|Z| < 2 \rightarrow$ Normal range.
- $|Z| \geq 2 \rightarrow$ Unusual.
- $|Z| \geq 3 \rightarrow$ Extremely unusual.

Here, $Z = 3$

This means the customer's spending is 3 standard deviations above the average, which is considered unusually high.

Answer : Yes. Since the z-score is 3, the customer's spending is unusually high compared with the average customer.

3. What would a negative z-score mean in this context?

A **negative z-score** means the customer's spending is **less than the average spending**.

For example:

If $Z = -1$

the customer spends **1 standard deviation below the average**.

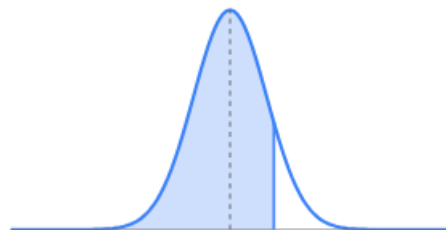
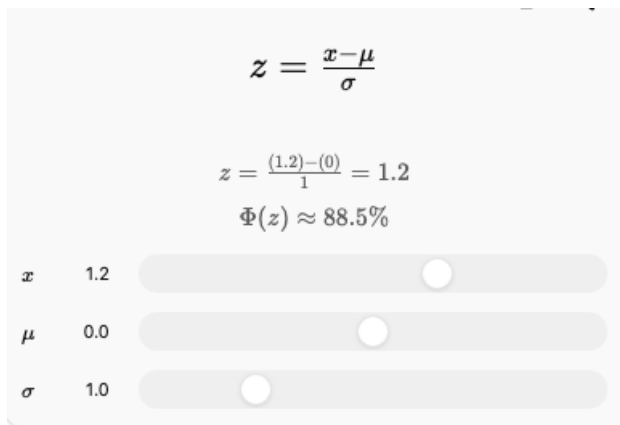
If, $Z = -2$

the customer spends **2 standard deviations below the average**, indicating much lower spending than most customers.

Answer

A **negative z-score** indicates that the customer spends **below the average amount**.

4. If another customer has a z-score of -1.5 , what does it tell you?



Given $Z = -1.5$

Interpretation

This means the customer's spending is 1.5 standard deviations below the average spending.

Since the average spending is ₹2000 and the standard deviation is ₹400:

Difference from the mean:

$$1.5 \times 400 = ₹600$$

Customer's spending:

$$2000 - 600 = ₹1400$$

Thus, the customer spends approximately ₹1400, which is below the average but not extremely unusual.

Answer

A z-score of -1.5 means the customer spends 1.5 standard deviations below the average, or approximately ₹1400. This indicates lower-than-average spending.

CASE STUDY 7 – Delivery Time Analysis

Given Data

Delivery times (minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Number of observations: $n = 10$

Q1. Calculate the Mean and Median

Step 1: Mean

Formula: Mean = $\frac{\sum x}{n}$

Calculate the sum

Value Running Sum

25	25
27	52
28	80
29	109
30	139
30	169

Value Running Sum

31 200

32 232

34 266

60 326

$$\sum x = 326$$

$$\text{Mean} = \frac{326}{10} = 32.6$$

Answer: Mean = 32.6 minutes

Step 2: Median

Since $n = 10$ (even),

$$\text{Median} = \frac{5^{th} + 6^{th}}{2}$$

5th value = 30

6th value = 30

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Answer: Median = 30 minutes

Q2. Find Q1 and Q3

Data:

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Lower half

25, 27, 28, 29, 30

Middle value = **28**

$$Q_1 = 28$$

Upper half

30, 31, 32, 34, 60

Middle value = **32**

$$Q_3 = 32$$

Answer: $Q_1 = 28$, $Q_3 = 32$

Q3. Calculate the IQR

Formula $IQR = Q_3 - Q_1$

$$= 32 - 28 = 4$$

Answer: $IQR = 4$

Q4. Check for Outliers Using the $1.5 \times IQR$ Rule

Step 1: Lower Limit

$$\begin{aligned} &Q_1 - 1.5(IQR) \\ &= 28 - 1.5(4) \\ &= 28 - 6 = 22 \end{aligned}$$

Step 2: Upper Limit

$$\begin{aligned} &Q_3 + 1.5(IQR) \\ &= 32 + 6 = 38 \end{aligned}$$

Acceptable range:

$$22 \leq x \leq 38$$

The value **60** is greater than 38.

Therefore,

60 is an outlier.

Q5. Explain Whether the Mean or Median Gives a Better Description

Mean = 32.6

Median = 30

The value **60** is an extreme value (outlier).

It increases the mean, while the median remains almost unchanged.

Therefore, the **median** gives a better description of the typical delivery time because it is not affected by the outlier.

Case Study 8 — Two Cities' Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

Given Data

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

Number of observations: $n = 7$

1. Calculate the mean temperature for each city.

City A

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculate the sum

Temperature Running Sum

28	28
29	57
30	87
30	117
31	148
30	178
32	210

$$\sum x = 210$$

Now,

$$\text{Mean} = \frac{210}{7} = 30^{\circ}\text{C}$$

Mean of City A = 30°C

City B

Calculate the sum

Temperature Running Sum

20	20
25	45
30	75
35	110

Temperature Running Sum

30	140
25	165
45	210

$$\sum x = 210$$

Now,

$$\text{Mean} = \frac{210}{7} = 30^{\circ}\text{C}$$

Mean of City B = 30°C

2. Calculate the variance for each city.

City A

Mean = 30

x	x - 30	(x - 30)²
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

$$\sum (x - \bar{x})^2 = 4 + 1 + 0 + 0 + 1 + 0 + 4 = 10$$

Variance

$$\sigma^2 = \frac{10}{7} = 1.43$$

Variance of City A = 1.43

City B

Mean = 30

x	x - 30	(x - 30) ²
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20	-10	100
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25	-5	25
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30	0	0
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35	5	25
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30	0	0
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25	-5	25
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45	15	225
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$$\sum (x - \bar{x})^2 = 100 + 25 + 0 + 25 + 0 + 25 + 225 = 400$$

Variance

$$\sigma^2 = \frac{400}{7} = 57.14$$

Variance of City B = 57.14

3. Which city has greater temperature variability?

To compare variability, calculate the standard deviation.

City A

$$\sigma = \sqrt{1.43} = 1.20^{\circ}C$$

City B

$$\sigma = \sqrt{57.14} = 7.56^{\circ}C$$

Since

$$7.56 > 1.20$$

Answer: City B has greater temperature variability because its standard deviation is much higher than City A's.

4. Why can two datasets have the same mean but different standard deviations?

To compare variability, calculate the standard deviation.

City A

$$\sigma = \sqrt{1.43} = 1.20^{\circ}C$$

City B

$$\sigma = \sqrt{57.14} = 7.56^{\circ}C$$

Since

$$7.56 > 1.20$$

Answer: City B has greater temperature variability because its standard deviation is much higher than City A's.

5. Which city has more predictable temperatures?

A smaller standard deviation means the temperatures are more consistent and predictable.

- City A: Standard deviation = 1.20°C
- City B: Standard deviation = 7.56°C

Since City A has a much smaller standard deviation, its temperatures are more consistent.

Answer: City A has more predictable temperatures because its standard deviation is much lower than that of City B.

CASE STUDY 9 – Data Center CPU Usage

Given Data

CPU Usage (%):

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Number of observations : $n = 10$

Q1. Calculate Mean, Median and Mode

Mean

Formula : Mean = $\frac{\sum x}{n}$

Sum

Value Running Sum

40	40
42	82
45	127
45	172
46	218
48	266
50	316
52	368
55	423
90	513

$$\sum x = 513$$

$$\text{Mean} = \frac{513}{10} = 51.3$$

Mean = 51.3%

Median :

$$\text{Median} = \frac{5^{th} + 6^{th}}{2}$$

5th value = 46

6th value = 48

$$= \frac{46 + 48}{2} = 47$$

Median = 47%

Mode :

45 occurs twice.

All others occur once.

Mode = 45%

Q2. Calculate the Range

Formula : Range = Maximum – Minimum

Maximum = 90

Minimum = 40

$$= 90 - 40 = 50$$

Answer: Range = 50%

Q3. Find Q1, Q3 and IQR

Data:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Lower Half

40, 42, 45, 45, 46

Middle value = **45**

$$Q_1 = 45$$

Upper Half

48, 50, 52, 55, 90

Middle value = **52**

$$Q_3 = 52$$

IQR

$$IQR = 52 - 45 = 7$$

Answers:

$$Q_1 = 45, Q_3 = 52, IQR = 7$$

Q4. Identify Possible Outliers

Lower Limit

$$45 - 1.5(7) = 45 - 10.5 = 34.5$$

Upper Limit

$$52 + 10.5 = 62.5$$

Acceptable range:

$$34.5 \leq x \leq 62.5$$

The value **90** is greater than 62.5.

Therefore,

90 is a possible outlier.

Q5. Determine the Direction of Skewness

Mean = 51.3

Median = 47

Since : Mean > Median

and there is a large value (**90**) on the right side,

the distribution is **positively skewed (right-skewed)**.

Case Study 10 — Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

1. Calculate the mean of both datasets.

Given Data

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

Number of observations:

Dataset A

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculate the sum

Value Running Sum

10 10

20 30

30 60

40 100

50 150

$$\sum x = 150$$

Now,

$$\text{Mean} = \frac{150}{5} = 30$$

Answer: Mean of Dataset A = 30

Dataset B

Calculate the sum

Value Running Sum

28 28

29 57

30 87

31 118

32 150

$$\sum x = 150$$

Now,

$$\text{Mean} = \frac{150}{5} = 30$$

Answer: Mean of Dataset B = 30

2. Calculate the variance of both.

Dataset A

Mean = 30

x	x - 30	(x - 30) ²
10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

$$\sum (x - \bar{x})^2 = 400 + 100 + 0 + 100 + 400 = 1000$$

Variance

Formula:

$$\begin{aligned}\sigma^2 &= \frac{\sum (x - \bar{x})^2}{n} \\ &= \frac{1000}{5} = 200\end{aligned}$$

Answer: Variance of Dataset A = 200

Dataset B

Mean = 30

x	x - 30	(x - 30) ²
28	-2	4
29	-1	1
30	0	0

x	x - 30	(x - 30) ²
31	1	1
32	2	4

$$\sum (x - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 = 10$$

Variance

$$= \frac{10}{5} = 2$$

Answer: Variance of Dataset B = 2

3. Calculate the standard deviation of both.

Formula

$$\sigma = \sqrt{\sigma^2}$$

Dataset A

Variance = 200

$$\begin{aligned}\sigma &= \sqrt{200} \\ &= 14.14\end{aligned}$$

Answer: Standard Deviation of Dataset A = 14.14

Dataset B

Variance = 2

$$\begin{aligned}\sigma &= \sqrt{2} \\ &= 1.41\end{aligned}$$

Answer: Standard Deviation of Dataset B = 1.41

4. Which dataset has greater spread?

The spread of a dataset is measured using its variance or standard deviation.

- Dataset A: Standard Deviation = 14.14
- Dataset B: Standard Deviation = 1.41

Since

$$14.14 > 1.41$$

Answer: Dataset A has the greater spread because its values are much more widely dispersed from the mean.

5. If both datasets represent model errors, which model appears more consistent?

A model is considered more consistent if its errors are closer to the mean, i.e., it has a smaller standard deviation.

- Dataset A: SD = 14.14

- Dataset B: SD = 1.41

Since Dataset B has the smaller standard deviation, its errors are much more consistent.

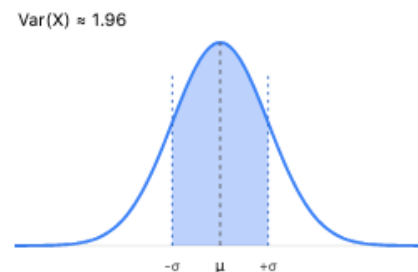
Answer: The model represented by Dataset B appears more consistent because it has a much lower standard deviation

6. Explain why mean alone is not enough to compare datasets.

$$Var(X) = E(X^2) - [E(X)]^2$$

$$Var(X) = (1.4)^2 = 1.96$$

σ 1.4



The **mean** tells us only the average value of a dataset. It does not indicate how the data values are distributed.

In this case:

- Both Dataset A and Dataset B have the **same mean of 30**.
- However, Dataset A values range from **10 to 50**, showing a large spread.
- Dataset B values range from **28 to 32**, showing a very small spread.

Therefore, although the means are identical, the variability is very different.

Answer: The **mean alone is not enough** because it does not measure the spread or consistency of the data. To compare datasets properly, **variance and standard deviation** must also be considered.